Homework 09

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All code is in Python (scikit-learn): `q1_svm_kernels.py` for Question 1 and `q2_svm_skill_analysis.py` for Question 2.

Question 1: Choosing an SVM kernel

Setup. The response is `chd` (n = 462; 160 cases, so the majority-class error is 0.346). The nine predictors are `sbp`, `tobacco`, `ldl`, `adiposity`, `famhist` (1 = Absent, 2 = Present), `typea`, `obesity`, `alcohol` and `age`. I split the observations into five folds myself, stratified by `chd` with seed 432 (fold sizes 93, 93, 92, 92, 92, each with 32 cases), and used the same folds for every candidate. In each fold a `StandardScaler` is fit on the four fitting folds only and then applied to the held-out fold.

Candidates (20 in total).

- Linear: C in {0.01, 0.1, 1} (3 settings).
- Polynomial, degrees 2, 3 and 4: C in {0.01, 0.1, 1} for each degree (9 settings). Held fixed: kernel $(\gamma\langle x, x'\rangle + r)^d$ with $\gamma = 1/p = 1/9$ and $r = 1$ (`coef0`).
- Radial: C in {1, 10} crossed with γ in {0.003, 0.01, 0.03, 0.1} (8 settings).

All fits use scikit-learn's `SVC` defaults otherwise: no class weights, and prediction by the sign of the decision function.

```
rng = np.random.default_rng(432)
fold = np.empty(n, dtype=int)
for cls in np.unique(y):                        # stratified 5 folds, built by hand
    idx = rng.permutation(np.flatnonzero(y == cls))
    fold[idx] = np.arange(idx.size) % 5

for name, params in candidates:
    errs = []
    for k in range(5):
        tr, te = fold != k, fold == k
        scaler = StandardScaler().fit(X[tr])  # fitting observations only
        model = SVC(**params).fit(scaler.transform(X[tr]), y[tr])
        errs.append(np.mean(model.predict(scaler.transform(X[te])) != y[te]))
```

Results. Mean 5-fold CV misclassification rate; SE = SD of the five fold errors / √5.

Kernel	C	γ	CV error	SE	Fold errors
linear	0.01		0.296	0.008	0.290, 0.323, 0.283, 0.304, 0.283
linear	**0.1**		**0.281**	**0.014**	**0.323, 0.280, 0.239, 0.272, 0.293**
linear	1		0.281	0.014	0.323, 0.280, 0.239, 0.272, 0.293
polynomial (d=2)	0.01		0.340	0.003	0.333, 0.333, 0.348, 0.348, 0.337
polynomial (d=2)	0.1		0.279	0.009	0.258, 0.290, 0.261, 0.283, 0.304
polynomial (d=2)	1		0.271	0.015	0.247, 0.312, 0.228, 0.283, 0.283

Kernel	C	γ	CV error	SE	Fold errors
polynomial (d=3)	0.01		0.327	0.007	0.301, 0.333, 0.337, 0.326, 0.337
polynomial (d=3)	0.1		0.268	0.012	0.247, 0.269, 0.239, 0.304, 0.283
polynomial (d=3)	1		0.288	0.008	0.269, 0.290, 0.272, 0.293, 0.315
polynomial (d=4)	0.01		0.290	0.004	0.280, 0.301, 0.283, 0.293, 0.293
polynomial (d=4)	0.1		0.268	0.011	0.237, 0.280, 0.250, 0.293, 0.283
polynomial (d=4)	1		0.301	0.014	0.301, 0.312, 0.272, 0.272, 0.348
radial	1	0.003	0.327	0.004	0.323, 0.323, 0.337, 0.315, 0.337
radial	1	0.01	0.275	0.008	0.269, 0.301, 0.250, 0.283, 0.272
radial	1	0.03	0.271	0.007	0.290, 0.258, 0.250, 0.283, 0.272
radial	1	0.1	0.290	0.011	0.258, 0.301, 0.272, 0.304, 0.315
radial	10	0.003	0.266	0.017	0.312, 0.280, 0.217, 0.239, 0.283
radial	10	0.01	0.271	0.006	0.269, 0.269, 0.250, 0.283, 0.283
radial	10	0.03	0.279	0.013	0.258, 0.301, 0.250, 0.272, 0.315
radial	10	0.1	0.305	0.011	0.290, 0.301, 0.293, 0.293, 0.348

The preferred setting is in bold.

Preferred model: linear kernel, C = 0.1. The lowest mean CV error is 0.266, from the radial kernel with C = 10 and γ = 0.003. It also has the largest SE (0.017), and its γ sits at the lower edge of the grid. Nearly every well-tuned candidate falls between 0.266 and 0.29: the best of each kernel are radial 0.266, polynomial d = 3 and d = 4 both 0.268, quadratic 0.271 and linear 0.281. These gaps are smaller than one standard error, so with 462 observations and five folds the data cannot tell the kernels apart. Under the one-standard-error rule (0.266 + 0.017 = 0.283), the linear SVM with C = 0.1 (0.281) qualifies. It is the simplest candidate: it has one tuning parameter instead of two or three, and its coefficients are interpretable. C = 0.1 and C = 1 give identical fold errors, so the choice is stable in C. The nonlinear kernels cut the CV error by at most about 1.5 percentage points, which does not justify their extra flexibility and tuning. Everything clearly worse is underfit: C = 0.01 for the polynomials and γ = 0.003 with C = 1 for the radial kernel are close to the majority-class error of 0.346.

Question 2: Write and use a short SVM analysis skill

I invoked the `svm-analysis` skill explicitly, giving the data details in the invocation rather than in the skill: *"Use the svm-analysis skill to analyze `saheart.csv`. The response is `chd`. Use the nine predictors `sbp`, `tobacco`, `ldl`, `adiposity`, `famhist`, `typea`, `obesity`, `alcohol` and `age`, coding `famhist` as 1 = Absent and 2 = Present."* The one-page, 11-point report it produced is on the next page, and the complete `SKILL.md` follows it.

Checks on the report and its calculations.

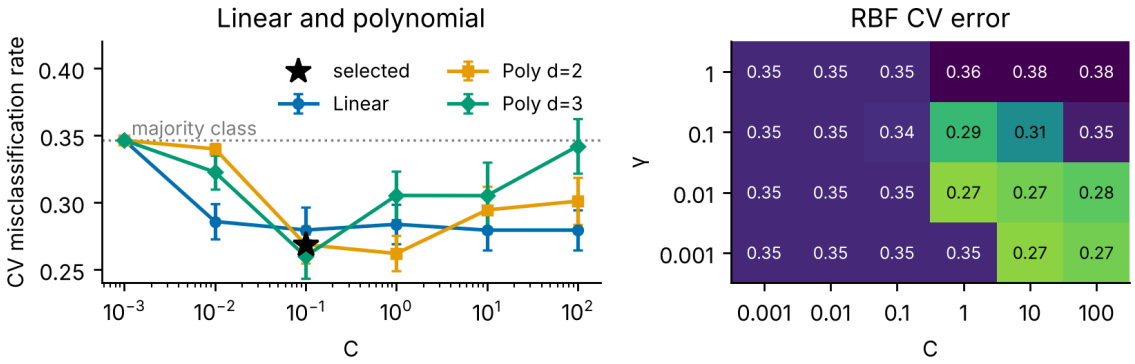
- The out-of-fold confusion matrix gives (30 + 94)/462 = 0.268 errors, which agrees with the mean fold error of 0.269. CHD recall is 66/160 = 0.41 and precision 66/96 = 0.69.
- The one-SE cutoff is 0.260 + 0.051/√10 = 0.276, so linear (0.279) is correctly excluded and quadratic with C = 0.1 (0.269) correctly included.

- Scaling is fit inside each fold by a pipeline, and every setting uses the same ten folds.
- The selected setting is not on the edge of its grid. Low RBF errors reach the $\gamma = 0.001$, $C = 10$ to 100 edge of the grid, which the report notes.
- This 10-fold, 36-setting analysis picks a different preferred model from Question 1 (quadratic vs. linear). In both analyses, the kernels differ by less than their standard errors.

SVM classification of coronary heart disease (South African heart disease data)

Data readiness. There are 462 men, 160 of them with CHD (34.6%), so the classes are moderately imbalanced and the majority-class error is 0.346. There are nine numeric predictors (`famhist` coded 1 = Absent, 2 = Present) and no missing values. The predictor SDs range from 0.49 to 24.5, so the predictors are standardized inside each CV fold. With $p \ll n$ and overlapping classes, a soft-margin SVM fits.

Kernels, tuning and selection. Kernels: linear (baseline), polynomial with d = 2 or 3 ($\gamma = 1/9$, coef0 = 1; risk-factor interactions) and RBF (general nonlinear). C ranges over 10^{-3} to 10^{2} and RBF γ over 10^{-3} to 1 (log grids; 36 settings). All settings use the same stratified 10-fold split (seed 432). The primary metric is the misclassification rate, with ROC-AUC (from the decision function) and F1 for CHD as secondary metrics. I chose the simplest setting (linear < quadratic < cubic < RBF, then smaller C) whose mean CV error is within one SE (SD/√10) of the lowest.



Kernel	Setting	CV error (mean ± SD)	ROC-AUC (mean ± SD)	F1
Polynomial d = 3	C = 0.1	0.260 ± 0.051	0.766 ± 0.069	0.513
Polynomial d = 2	C = 1	0.262 ± 0.042	0.762 ± 0.059	0.528
Polynomial d = 2	**C = 0.1 (selected)**	**0.269 ± 0.044**	**0.788 ± 0.079**	**0.504**
RBF	C = 1, γ = 0.01	0.269 ± 0.045	0.782 ± 0.080	0.519
Linear	C = 0.1	0.279 ± 0.054	0.784 ± 0.081	0.548
Majority class	—	0.346	0.500	0

Selected model and fit. The lowest CV error is 0.260 (cubic, C = 0.1), so the one-SE cutoff is 0.276. The linear kernel (0.279) falls just outside it, and the quadratic kernel with C = 0.1 is the simplest setting inside. Refit on all 462 observations, it has 293 support vectors (63% of n; 149 without and 144 with CHD). 270 of them sit at the bound $\alpha = C$, and 262 observations lie inside the margin, whose width $2/\|w\|$ is 1.05 in standardized feature-space units. Training error is 0.242 against CV error 0.269, so overfitting is small; larger C does overfit (0.30 at C = 100). From out-of-fold predictions, it correctly classifies 272 of 302 non-cases and 66 of 160 cases. CHD recall is 0.41 and precision 0.69; the no-CHD values are 0.90 and 0.74.

Synthesis and limitations.

- The SVM cuts the error from 0.346 to about 0.27. The well-tuned kernels differ by under 2 points, well within the fold SDs of about 0.05, and they all reach an AUC of about 0.78.
- Most observations are support vectors: the boundary rests on weak, noisy separation.
- With the default threshold, cases are under-detected (recall 0.41).
- There is no separate test set, so the selected CV error is slightly optimistic. The RBF optimum is near the edge of its grid (small γ, large C), a nearly linear regime.

Appendix: complete `SKILL.md`

```
---
name: svm-analysis
description: Guidance for evaluating data suitability, kernel choice, hyperparameter
tuning, model selection, and reporting for Support Vector Machines (SVM).
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1. Data Suitability Assessment
   - Verify sample size ($N$), feature dimension ($p$), feature types, scaling, and class
balance.
   - Confirm linear/non-linear separability potential and check for severe missingness or
excessive noise.

2. Kernel Selection & Justification
   - Linear Kernel: Default baseline, ideal for high-dimensional data ($p \gg N$) or
linearly separable boundaries.
   - Radial Basis Function (RBF) Kernel: General non-linear boundary standard; flexible
for complex patterns.
   - Polynomial Kernel: Suitable when specific interaction degrees between features are
hypothesized.

3. Hyperparameter Tuning & Strategy
   - Parameters: Regularization ($C$), kernel scale ($\gamma$ for RBF), polynomial degree
($d$).
   - Search Method: Stratified $k$-fold cross-validation using grid or randomized search
over log-scaled grids.

4. Evaluation & Final Model Selection
   - Select primary metrics based on balance (e.g., Accuracy, F1-Score, ROC-AUC, Log-
Loss).
   - Use cross-validation performance distributions to select the final hyperparameter
configuration.

5. Visual & Tabular Artifacts
   - Produce one multi-panel figure showing decision boundaries or hyperparameter search
heatmaps.
   - Provide one compact table summarizing candidate models, hyperparameters, and CV
metrics ($\mu \pm \sigma$).

6. Fit Interpretation & Key Results
   - Report support vector count, margin size, training vs. validation performance, and
key metrics.
   - Detail confusion matrix results, class-wise precision/recall, and potential
overfitting signs.

7. Concise Synthesis (1-Page Limit at 11pt)
   - Limit report length to under ~500 words to guarantee a single-page fit.
   - Prioritize evidence hierarchy: Data readiness $\rightarrow$ Tuning landscape
$\rightarrow$ Final performance metrics.
```